

Artificial Intelligence & Machine Learning Internship

Task 2: Feature Engineering, Model Optimization & Performance Comparison

Abstract

This project demonstrates an end-to-end machine learning workflow using the California Housing dataset. The notebook covers preprocessing, feature scaling, model training, evaluation, comparison, and visualization.

Objectives

- Understand feature engineering and preprocessing.
- Train multiple regression models.
- Compare models using RMSE and R^2 .
- Visualize prediction performance.

Dataset

The California Housing dataset from scikit-learn was used. Features describing housing characteristics were used to predict median house value.

Tools & Libraries

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib and Scikit-learn.

Methodology

1. Load and inspect the dataset.
2. Separate features and target.
3. Split into training and testing sets.
4. Apply StandardScaler to numerical features.
5. Train Linear Regression, Ridge Regression and Decision Tree Regressor.
6. Generate predictions.
7. Evaluate models using RMSE and R^2 .
8. Compare results and visualize Actual vs Predicted values.

Models

Linear Regression provides a strong baseline. Ridge Regression helps reduce overfitting through regularization. Decision Tree Regressor captures non-linear relationships and offers an alternative approach for comparison.

Results

The trained models were evaluated using Root Mean Squared Error (RMSE) and R^2 Score. Performance comparison helped identify the most suitable model for the dataset. Visualization of Actual vs Predicted values provided additional insight into prediction quality.

Learning Outcomes

- Data preprocessing
- Feature engineering
- Feature scaling
- Regression algorithms
- Model comparison
- Performance evaluation
- Data visualization

Conclusion

The project successfully implemented the complete regression workflow. It strengthened practical skills in data preparation, model development, evaluation, and performance comparison using Python and Scikit-learn.